MAURICE FILO

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GOAL

My interdisciplinary background in electrical and mechanical engineering, systems biology, and synthetic biology is unified by a focus on control and dynamical systems theory. My goals are threefold: (1) to advance the theoretical foundations of control systems, with an emphasis on robust control in stochastic environments; (2) to develop computational and theoretical tools for the analysis and design of control systems in genetic circuits and ear biomechanics; and (3) to explore the use of reinforcement learning in genetic circuit design where traditional theoretical methods fall short.

EDUCATION

Postdoctoral Researcher, Control Theory and Systems Biology (CTSB), Prof. Mustafa Khammash

Eidgenössische Technische Hochschule (ETH) Zürich

2018-2023 | Basel, Switzerland

Ph.D. & Masters in Mechanical Engineering, Prof. Bassam Bamieh

University of California, Santa Barbara (UCSB)

2013-2018 | California, USA

- Ph.D. Dissertation: Topics in Stochastic Stability, Optimal Control & Estimation Theory
- Master Thesis: Topics in Modeling of Cochlear Dynamics: Computation, Response & Stability Analysis
- Majors: Dynamic Systems, Control & Robotics (DSCR), Computational Science & Engineering (CSE) — GPA: 4/4

Masters in Electrical & Computer Engineering

American University of Beirut (AUB)

2010-2013 | Beirut, Lebanon

- Master Thesis: Nonlinear Nonlocal Two Dimensional Cochlear Modeling
- Majors: Control Theory, Signal & Image Processing — GPA: 4/4

Diploma in Electrical Engineering

Lebanese University (LU)

2005-2010 | Beirut, Lebanon

- Diploma Thesis: Demining Robot Vision
- Majors: Control & Power Systems — First-Class Honours

FELLOWSHIPS & AWARDS

Ranked First in the 2023 Competition for the Research Officer Position at INRIA, France (Declined by Applicant)
https://www.inria.fr/sites/default/files/2023-06/Resultats_Results_admission_ISFP_2023_2.pdf

"PID Feedback Control Systems for Robust Control of Molecular Networks" ranked among the Top 20 Inventions
Spark Award (2021), https://ethz.ch/en/industry/researchers/ip/sparkaward/2021.html

Best PhD Thesis Award
Center for Control, Dynamical-Systems & Computations (CCDC) (2019)

Featured as the Researcher of the Term
Center for Control, Dynamical-Systems & Computations (CCDC) (2016)
https://www.ccdc.ucsb.edu/content/maurice-filo

Best Teacher Assistant Award
Department of Mechanical Engineering, UCSB (2015)

CCDC Fellowship
Center for Control, Dynamical-Systems & Computations (CCDC) (2013)

Holbrook Foundation Fellowship
Institute for Energy Efficiency (IEE), UCSB (2013)

RESEARCH PUBLICATIONS & PATENTS

Journal Articles

- **M. Filo** and B. Bamieh, “An input–output approach to structured stochastic uncertainty in continuous time,” *International Journal of Robust and Nonlinear Control*, vol. 35, no. 11, pp. 4462–4479, 2025.
- **M. Filo***, S. Aoki*, M. Hou*, S. Anastassov, and M. Khammash, “Engineering genetic controllers to accelerate adaptation and attenuate cellular noise,” *under review in Cell Systems*, 2025.
- S. Anastassov, **M. Filo**, and M. Khammash, “Inteins: A swiss army knife for synthetic biology,” *Biotechnology Advances*, p. 108 349, 2024.
- **M. Filo***, A. Gupta*, and M. Khammash, “Anti-windup strategies for biomolecular control systems facilitated by model reduction theory for sequestration networks,” *Science Advances*, vol. 10, no. 34, eadl5439, 2024.
- S. Anastassov*, **M. Filo***, C.-H. Chang, and M. Khammash, “A cybergenetic framework for engineering intein-mediated integral feedback control systems,” *Nature Communications*, vol. 14, no. 1, p. 1337, 2023.
- **M. Filo**, C.-H. Chang, and M. Khammash, “Biomolecular feedback controllers: From theory to applications,” *Current Opinion in Biotechnology*, vol. 79, p. 102 882, 2023.
- **M. Filo**, S. Kumar, and M. Khammash, “A hierarchy of biomolecular proportional-integral-derivative feedback controllers for robust perfect adaptation and dynamic performance,” *Nature Communications*, vol. 13, no. 1, pp. 1–19, 2022.
- T. Frei, C.-H. Chang, **M. Filo**, A. Arampatzis, and M. Khammash, “A genetic mammalian proportional–integral feedback control circuit for robust and precise gene regulation,” *Proceedings of the National Academy of Sciences*, vol. 119, no. 00, e2122132119, 2022.
- B. Bamieh and **M. Filo**†, “An input–output approach to structured stochastic uncertainty,” *IEEE Transactions on Automatic Control*, vol. 65, no. 12, pp. 5012–5027, 2020.
- **M. Filo**† and B. Bamieh, “Investigating instabilities in the mammalian cochlea using a stochastic uncertainty model,” *IEEE Transactions on Molecular, Biological and Multi-Scale Communications*, vol. 6, no. 1, pp. 1–12, 2020.
- N. Hajj, **M. Filo**, and M. Awad, “Automated composer recognition for multi-voice piano compositions using rhythmic features, n-grams and modified cortical algorithms,” *Complex & Intelligent Systems*, vol. 4, no. 1, pp. 55–65, 2018.
- **M. Filo**, F. Karameh, and M. Awad, “Order reduction and efficient implementation of nonlinear nonlocal cochlear response models,” *Biological cybernetics*, vol. 110, no. 6, pp. 435–454, 2016.

Conference Proceedings

- **M. Filo** and M. Khammash, “Realizing reduced and sparse biochemical reaction networks from dynamics,” in *2025 IEEE 64th Conference on Decision and Control (CDC) [to appear]*, IEEE, 2025. arXiv: 2508.18096. [Online]. Available: https://arxiv.org/abs/2508.18096.
- **M. Filo**, S. Kumar, S. Anastassov, and M. Khammash, “Exploiting the nonlinear structure of the antithetic integral controller to enhance dynamic performance,” in *2022 IEEE 61st Conference on Decision and Control (CDC)*, IEEE, 2022, pp. 1294–1299.
- **M. Filo** and M. Khammash, “Optimal parameter tuning of feedback controllers with application to biomolecular antithetic integral control,” in *2019 IEEE 58th Conference on Decision and Control (CDC)*, IEEE, 2019, pp. 951–957.
- **M. Filo** and B. Bamieh, “A block diagram approach to stochastic calculus with application to multiplicative uncertainty analysis,” in *2018 IEEE Conference on Decision and Control (CDC)*, IEEE, 2018, pp. 3270–3275.
- **M. Filo** and B. Bamieh, “Function space approach for gradient descent in optimal control,” in *2018 Annual American Control Conference (ACC)*, IEEE, 2018, pp. 3447–3453.
- **M. Filo** and B. Bamieh, “Investigating cochlear instabilities using structured stochastic uncertainty,” in *2017 IEEE 56th Annual Conference on Decision and Control (CDC)*, IEEE, 2017, pp. 1634–1640.

- **M. Filo** and B. Bamieh, "Sensor motion for optimal estimation in distributed dynamic environments," in *2017 American Control Conference (ACC)*, IEEE, 2017, pp. 3263–3269.

Patents

- **M. Filo**, S. Anastassov, C.-H. Chang, and M. Khammash, "Intein-based controllers," Patent pending.
- T. Frei, C.-H. Chang, **M. Filo**, and M. Khammash, "Pid feedback control systems for robust control of molecular networks," Patent pending.

CONTRIBUTED TALKS & CONFERENCE SESSIONS

Engineering Genetic Controllers to Accelerate Adaptation and Attenuate Cellular Noise
2024 CUED Control Group Seminars, University of Cambridge, UK

Organized and chaired an invited conference session on Biological Controllers
2022 Conference on Decision and Control (CDC) in Cancun, Mexico

Distributed Structured Stochastic Uncertainties, Cochlear Instabilities
26th International Symposium on Mathematical Theory of Networks and Systems in Cambridge, UK

Organized and chaired an invited conference session on Biological Controllers
2022 Conference on Decision and Control (CDC) in Cancun, Mexico

Exploiting the Nonlinear Structure of the Antithetic Integral Controller to Enhance Dynamic Performance
2022 Conference on Decision and Control (CDC) in Cancun, Mexico

Optimal Parameter Tuning of Feedback Controllers with Application to Biomolecular Antithetic Integral Control
2019 Conference on Decision and Control (CDC) in Nice, France

A Block Diagram Approach to Stochastic Calculus with Application to Multiplicative Uncertainty Analysis
2018 Conference on Decision and Control (CDC) in Miami, USA

A Function Space Approach to Gradient Descent in Optimal Control
2018 American Control Conference (ACC) in Wisconsin, USA

Investigating Cochlear Instabilities Using Structured Stochastic Uncertainty
2017 Conference on Decision and Control (CDC) in Melbourne, Australia

Sensor Motion for Optimal Estimation in Distributed Dynamic Environments
2017 American Control Conference (ACC) in Seattle, Washington

Possible Sources of Instabilities in the Cochlea
2015 SIAM Conference on Applications of Dynamical Systems in Snowbird, Utah

RESEARCH EXPERIENCE

Postdoc & Senior Research Scientist

Eidgenössische Technische Hochschule (ETH) Zürich

2018 – present | Basel, Switzerland

- Currently developing a **deep machine learning** approach to design reaction networks with pre-specified properties.
- Led a subgroup of biologists and engineers to design, model, analyze, and test advanced **genetic control systems**.
- Designed and analyzed a series of biomolecular **Proportional-Integral-Derivative (PID)** controllers (patent pending).
- Established a unifying **model analysis and reduction theory** for intein-based controllers (patent pending).
- Designed molecular reaction networks capable of realizing **anti-windup** schemes.
- Studied **the impact of resource burden on genetic circuits** and developed guidelines to minimize their effects.

* co-first author, † corresponding author

- Designed and implemented a numerical method for the **optimally tuning parameters** of nonlinear feedback controllers.
- Developed a user-friendly **MATLAB application** for modeling and simulating biomolecular controllers in both deterministic and stochastic environments (youtube link: https://www.youtube.com/watch?v=SeFFJz4dxOA).
- Served as a review editor and reviewer for multiple journals and conferences.
- Co-authored and co-conceived **two successfully awarded proposals**—SNSF Advanced Grant (216505) and ERC Advanced Grant (2024), lead by Prof. Mustafa Khammash as PI.

Graduate Research Assistant

University of California, Santa Barbara (UCSB)

2013 – 2018 California, USA

- Developed a mathematical framework for **optimal field estimation via mobile tomographic sensors**.
- Designed and implemented a new numerical method for solving **optimal control** problems.
- Developed a framework for studying **stochastic stability** of dynamical systems with stochastic uncertainties.
- Developed a stochastic **biomechanical model for the cochlea** with connections to otoacoustic emissions and tinnitus.
- Devised multiple numerical techniques for **efficient simulations** of cochlear models.
- Studied mean-square stability in **infinite-dimensional** stochastic dynamical systems.

Graduate Research Assistant

American University of Beirut (AUB)

2011 – 2013 Beirut, Lebanon

- Designed, implemented and analyzed an **efficient model reduction** scheme for the cochlea.
- Designed and implemented a **deep machine learning algorithm** for capturing the style of **piano composers**.

TEACHING EXPERIENCE

Lecturing & Lab Tutoring

Ashesi University

2023, 2024 Ghana, Africa

- Participated in the ETH for Development (ETH4D) Master program in Mechatronics.
- Assisted in developing course materials for advanced control systems design and analysis.
- Delivered lectures and managed exercise and office hours sessions.
- Prepared and conducted hands-on tutorials on robotic programming of control algorithms, including LQR and LQG.

Lecturing & Graduate Students Supervision

Eidgenössische Technische Hochschule (ETH) Zürich

2018-present Basel, Switzerland

- Introduction to Dynamical Systems with Professor Mustafa Khammash
- Advanced Bioengineering with Professor Mustafa Khammash
- Computational Biology & Bioinformatics Seminar
- Co-supervision of 3 Master Students (1 Master Thesis and 2 Master Projects)
- Co-supervision of 2 Doctoral Students

Teaching Assistant

University of California, Santa Barbara (UCSB)

2013 – 2015 California, USA

- Mechatronics lab instructor with Professor Bassam Bamieh
- Vibrations with Professor Igor Mezic
- Control Systems with Professor Bradley Paden

Teaching Assistant

American University of Beirut (AUB)

2011 – 2013 Beirut Lebanon

- Graduate course in Pattern Recognition with Professor Mariette Awad
- C++ lab instructor
- Electronics lab instructor

FIELD WORK EXPERIENCE

Electrical Engineer Intern

National Oilwell Varco (NOV)

Summer 2009 Dubai, UAE

- Troubleshooting of faults in Silicon Controlled Rectifier (SCR) rooms.
- Complete reassembling of SCR units for maintenance.

Electrical Engineer Intern

Electricité De Zahle (EDZ)

Summer 2008 Zahle, Lebanon

- Maintenance of power grids.
- Installations of new power transformers and transmission lines.
- Development of software power grids in Zahle using Geographic Information System (GIS).

PROGRAMMING & SOFTWARE SKILLS

MATLAB | Simulink | Simscape | LabVIEW | C | C++ | Python | PyTorch | Julia | LaTeX

PIC | PLC | SBRIO | NICVS

Blender | TikZ | Adobe Illustrator

EXTRACURRICULAR EXPERIENCE

Private Piano Teacher

2004 – 2013 Lebanon

Music Major, Lebanese National Conservatory

1993 – 2005 Zahle & Beirut, Lebanon

- Piano, Music Theory, Solfegio and Harmony.

Classical Piano Concert

April, 2004 Grand Kadri Hotel, Lebanon